The Holonet Machine

A complete engineering blueprint for a computer
whose wiring diagram is a piece of finite geometry

Every figure in this document is measured or proved.

W(3,3)–E_8 project, glue track

3 August 2026 · Passes 2700–2837

Abstract

This is a build document for a computer that does not exist yet, every part of which has been either mathematically proved, formally verified, or physically measured by someone. The machine's logic is a 40-point finite geometry called $W(3,3)$. Its public instruction set has eight opcodes in three bits; its internal frame engine needs only *four* operations in two bits, and both generate the same group of order 4,199,040 exactly. The public unit fits in **72 logic cells** on a $25 FPGA at 60.8 MHz; the minimal engine fits in **43** at **72.4 MHz** — both measured, neither inferred from the other. Its physical layer is a photonic gate a laboratory has already demonstrated at 0.92 fidelity, and its magic-state resource now has an explicit fidelity-improving protocol on the deep grade. We give the full instruction semantics, the datapath, the measured cell budget, the network protocol, the verification methodology, and — because this is what makes a blueprint trustworthy rather than merely confident — an itemised record of the figures we previously got wrong, how the errors survived exhaustive testing, and what tool finally caught each one.

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1 How to read this document

This document has two readers in mind and does not compromise for either.

If you are not an engineer.
Read the cream boxes and the diagrams. They are complete: every idea in the machine is explained there from scratch, in order, with no symbols you have not already met. You may skip every blue box without losing the thread.

If you are an engineer.
The blue boxes carry the specification: exact semantics, matrices, cell counts, clock frequencies, proof obligations and the commands that reproduce them. The cream boxes are there so you can hand this to a colleague in a different field.

A third kind of box appears throughout:

What we got wrong, and how we know. These record something this project believed, published, and then had to withdraw — with the measurement that overturned it. There are eleven of them. They are not apologies; they are the reason the other numbers in this document can be trusted. A blueprint with no errata is a blueprint nobody has checked.

2 The machine in one page

In plain language. *Every computer you have used stores information as* bits *— switches that are either off or on. This machine stores information in* qutrits*: units with three settings instead of two. That sounds like a small change. It is not, because of what the three settings let you do with* geometry.

Here is the strange and central fact. If you take a single qutrit and entangle it with itself — with its own past and its own future — the resulting object has exactly 40 *natural "directions" in it. Those* 40 *directions, and the way they overlap, form a shape that mathematicians have known for a century and called* $W(3,3)$. *The machine we are describing does not* use *that shape as a convenience. The shape* is *the machine: its memory layout, its instruction set, its network topology, and its error model are all readings of the same* 40*-point object.*

One qutrit, entangled with its own past, is enough to generate the entire substrate. The 40 points of $W(3,3)$ are not 40 things — they are 40 views of one thing.

The machine, specified.

Logical substrate	$W(3,3)$: the symplectic generalised quadrangle of order $(3,3)$. 40 points, 40 totally isotropic lines, collinearity graph $\mathrm{SRG}(40,12,2,4)$.
State	One qutrit pair (past, future) living in $\mathbb{C}^9 = \mathrm{End}(\mathbb{C}^3)$.
Tracked quantity	The *Pauli frame*: four trits $(x_p, z_p, x_f, z_f) \in \mathbb{F}_3^4$, i.e. 81 states.
Public ISA	$\mathcal{I}_{\mathrm{holo}}$, eight opcodes, three bits.
Internal micro-ISA	F_p, $\mathrm{CX}_{p\to f}$, $\mathrm{CX}_{f\to p}$, Z_p — four operations, two bits, no illegal encoding.
Group generated	$\mathrm{ASp}(4,3) = \mathbb{F}_3^4 \rtimes \mathrm{Sp}(4,3)$, order $81 \times 51840 = 4{,}199{,}040$ *exactly.*
Arithmetic unit	Public: 72 iCE40 cells at 60.80 MHz. Minimal: **43** cells at **72.40** MHz. Both measured, §6.2.
Physical layer	Time-bin × frequency-bin photonic qutrits; the two-qutrit SUM gate is experimentally demonstrated at fidelity 0.92 ± 0.01.
Network	40-ary fractal; $N_n = 40^n$ leaves, routing depth $8n = 8\log_{40} N$; the routing code satisfies Kraft equality exactly, so *every* bit string is a legal address.

The whole system, at a glance

NETWORK — 40-ary fractal holonet; remote SUM costs one Bell pair + two classical trits §11

INSTRUCTION SET — eight opcodes; generates $\mathrm{ASp}(4,3)$, order 4,199,040 §5

FRAME UNIT — 4 trits of state, 72 logic cells, 60.80 MHz §6

GEOMETRY — $W(3,3)$: 40 points, 40 lines, $\mathrm{SRG}(40,12,2,4)$ §4

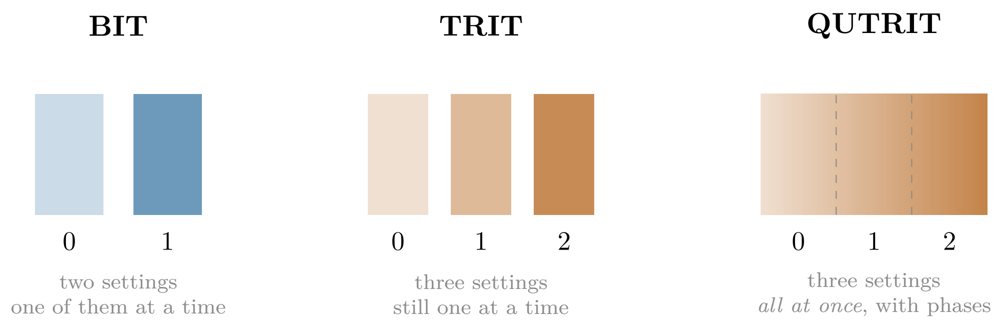
PHYSICS — one self-entangled qutrit; time bins × frequency bins §3

The unusual property of this stack is that the layers are not independent design choices stacked by convention. Each one is *forced* by the one below it. The network is 40-ary because the geometry has 40 points. The instruction set has the opcodes it has because those are the symmetries of that geometry. This is what the rest of the document demonstrates, layer by layer.

3 Qutrits, and what "self-entangled" means

3.1 From bits to qutrits

In plain language. *A bit is a switch: off or on,* 0 *or* 1*. A* trit *is a three-way switch:* 0, 1, *or* 2*. Ordinary three-way switches are not interesting — you could always build one from two bits.*
A qutrit *is different. It is a physical three-way switch obeying quantum mechanics, which means it can be in a blend of all three settings at once, and — this is the part that matters —* two *qutrits can be correlated in ways that no pair of classical switches can imitate.*



Exactly. A qutrit is a unit vector in $\mathbb{C}^3$. Two qutrits live in $\mathbb{C}^3 \otimes \mathbb{C}^3 = \mathbb{C}^9$. The relevant operators are generated by

$$X\,|j\rangle = |j+1 \bmod 3\rangle\,, \qquad Z\,|j\rangle = \omega^j\,|j\rangle\,, \qquad \omega = e^{2\pi i/3},$$

which satisfy $ZX = \omega XZ$. Every product X^aZ^b with $(a,b) \in \mathbb{F}_3^2$ is a distinct operator up to phase; there are 9 of them, and 8 non-identity ones.

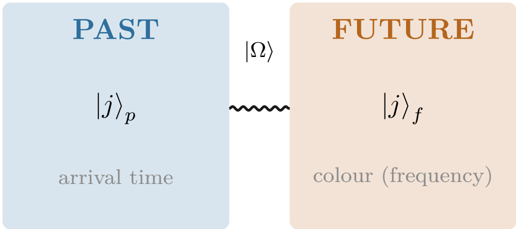
3.2 The self-entanglement, and where the 40 comes from

In plain language. *Now the key move. Take* one *qutrit and entangle it with itself at two different times — call them "past" and "future". Physically this is done by splitting a single photon's degrees of freedom: its arrival* time *carries one qutrit and its* colour *carries the other. It is one particle, entangled with its own history.*
Mathematically, an operator acting on one qutrit is a 3×3 *table of numbers, and there are* 9 *independent slots in such a table. So the space of things you can* do *to one qutrit is itself* 9*-dimensional — exactly the size of the space of* states *of two qutrits. This is not a coincidence or an analogy. It is an identity:*

$$\underbrace{\mathbb{C}^3 \otimes \mathbb{C}^3}_{\textit{two qutrits}} = \underbrace{\mathrm{End}(\mathbb{C}^3)}_{\textit{operations on one qutrit}}.$$

So "the two-qutrit machine" and "the one-qutrit machine that can act on itself" are the same machine described twice.
Now count the directions. There are 81 *ways to label a pair (four trits), one of which is "do nothing". That leaves* 80*. Each direction and its opposite behave identically for our purposes, and each direction comes in* 2 *such flavours, so the distinct directions number* $80/2 = 40$.

$$\frac{3^4-1}{2} = \frac{81-1}{2} = 40.$$ **The forty points of the geometry are the forty independent things one self-entangled qutrit can be measured along.**



$$|\Omega\rangle = \mathrm{CX}_{p\to f}\,(F_3 \otimes I)\,|0\rangle_p\,|0\rangle_f = \frac{1}{\sqrt{3}}\sum_{j=0}^{2}|j\rangle_p\,|j\rangle_f$$

One photon. Two degrees of freedom. The entanglement is between the particle and itself.

What we got wrong, and how we know. The sentence "one qutrit, self-entangled as past/future, is enough to generate the full substrate" is *not* ours. It is stated verbatim in this project's own encyclopedia, `docs/index.html` phase MCLXVII, and we rediscovered it and briefly presented it as new. What survives as new is narrower: the explicit construction of the 40 points as projective left×right multiplication classes on $\mathrm{End}(\mathbb{C}^3)$, and the identification of the 40 lines as the maximal *commuting* subalgebras. The lesson — search for the *result*, not the topic — produced three of the tools in §12.

4 The substrate: $W(3,3)$

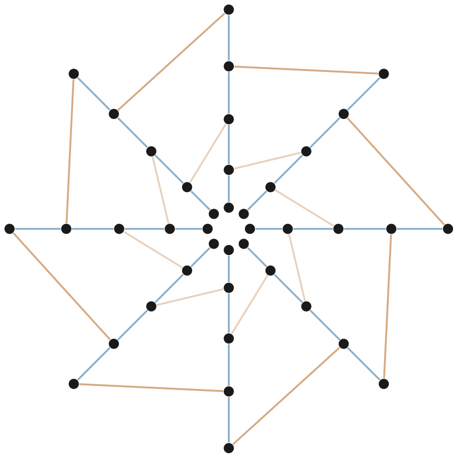
In plain language. *Here is what the* 40 *directions look like when you write down which of them are compatible with which.*
Two measurements are compatible *if you can perform both without one disturbing the other. Of the* 40 *directions, each is compatible with exactly* 12 *others. Whenever two directions are compatible, exactly* 2 *further directions are compatible with both. Whenever two are* in*compatible, exactly* 4 *are compatible with both.*
Those four numbers — 40, 12, 2, 4 — pin the shape down completely. There is essentially one object in the world with that compatibility pattern, and it has been studied since the 1900s under the name $W(3,3)$.

Exactly. $W(3,3)$ is the symplectic generalised quadrangle of order $(3,3)$: the points are the 40 points of $\mathrm{PG}(3,3)$ and the lines are the 40 totally isotropic lines of a non-degenerate alternating form. Its collinearity graph is strongly regular with parameters

$$\mathrm{SRG}(40,\ 12,\ 2,\ 4),$$

i.e. 40 vertices, every vertex of degree 12, adjacent pairs with $\lambda = 2$ common neighbours, non-adjacent pairs with $\mu = 4$. Its automorphism group is $\mathrm{PGSp}(4,3) = U_4(2){:}2 = W(E_6)$ of order 51,840.

4.1 Why this shape and not another



40 points, 40 lines.

Every point lies on exactly 4 lines.
Every line carries exactly 4 points.
Given a line ℓ and a point $P \notin \ell$, there is *exactly one* point of ℓ collinear with P.

(Schematic. The full incidence graph has 160 flags and does not embed in the plane without crossings; this figure shows the ring structure and two of the line families only.)

In plain language. *That last property — "exactly one" — is the entire content of the word* quadrangle. *It is a strong constraint: it means the geometry contains no triangles, so there is never any ambiguity about how to get from one place to another. For a computer, that is precisely what you want from an address space and from a routing fabric: there is one shortest path, and no loops to break ties in.*

4.2 The two groups of order 51,840, which are not the same group

Exactly. Two distinct groups of order 51,840 appear in this project, and confusing them is the most common error in this area:

$$\mathrm{Sp}(4,3) = 2.U_4(2) \quad \text{(centre of order 2)}, \qquad \mathrm{PGSp}(4,3) = U_4(2){:}2 = W(E_6) \quad \text{(trivial centre)}.$$

They are *not* isomorphic. The first is the group acting on Pauli frames — the one the hardware implements. The second is the automorphism group of the geometry — the one the drawing has. One is a central double cover; the other is an outer extension.

In plain language. *This distinction is not pedantry, and it has physical content. The extra element in the first group acts as a sign, $z \mapsto -1$. Whether that sign is "spent" or "free" decides whether the resulting structure has a handedness — whether it distinguishes left from right. This is why the machine can* host *a chirality, though as we established in earlier work it cannot* explain *one from the inside.*

5 The instruction set

5.1 The idea of a Pauli frame

In plain language. *Here is the trick that makes the machine cheap.*
A quantum state is expensive to write down: for two qutrits you need nine complex numbers, and keeping them accurate is most of what makes quantum computing hard. But for a large and useful family of operations — the Clifford *operations — you never need the state at all. You only need a* label *saying how the state has been shifted relative to where it started.*

That label is four trits. Four trits is eight bits. The machine's entire "register file" for tracking Clifford operations is one byte.

The frame unit tracks a label, not a state. Four trits, 81 possible values, eight bits of memory — and it correctly accounts for a group of 4,199,040 distinct operations.

Exactly. The Pauli frame is $(x_p, z_p, x_f, z_f) \in \mathbb{F}_3^4$, denoting the operator $X^{x_p}Z^{z_p} \otimes X^{x_f}Z^{z_f}$. A Clifford unitary U acts on frames by conjugation, $U(X^aZ^b)U^\dagger = X^{a'}Z^{b'}$ up to phase, and the induced map on $\mathbb{F}_3^4$ is *linear and symplectic*: it preserves

$$\langle u, v\rangle \;=\; (x_p z_p' - z_p x_p') + (x_f z_f' - z_f x_f') \bmod 3.$$

5.2 The eight opcodes

$\mathcal{I}_{\text{holo}}$**, complete semantics.**

Op	**Name**	**Action on** (x_p, z_p, x_f, z_f)	**Kind**
`000`	F_p	$(x_p, z_p) \mapsto (-z_p,\ x_p)$	linear
`001`	F_f	$(x_f, z_f) \mapsto (-z_f,\ x_f)$	linear
`010`	S_p	$z_p \mapsto z_p + x_p$	linear
`011`	S_f	$z_f \mapsto z_f + x_f$	linear
`100`	CX	$d{=}0$: $z_p \mapsto z_p - z_f$, $x_f \mapsto x_f + x_p$	linear
		$d{=}1$: $x_p \mapsto x_p + x_f$, $z_f \mapsto z_f - z_p$	linear
`101`	$\sigma^5{=}Z$	$z_p \mapsto z_p + 1$ or $z_f \mapsto z_f + 1$	**affine**
`110`	D_{12}-mirror	transport in the dihedral group D_{12}	(not a frame op)
`111`	M_{36}-magic	typed request for one of 36 Witting rays	(handshake)

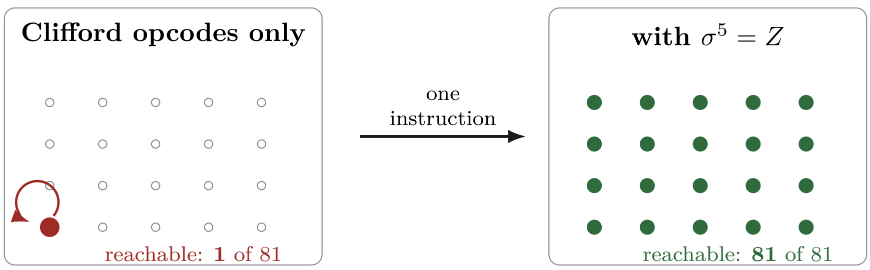
5.3 The reachability theorem, and why it matters

In plain language. *Now a question that sounds trivial and is not:* starting from a blank machine, which states can these instructions actually reach?

The answer is startling. All six of the Clifford instructions, applied in any order, any number of times, reach exactly one *state — the blank one. They cannot move the machine at all.*

The reason is simple once you see it. Those six instructions are all linear, *and a linear map always leaves the origin where it is. Multiplying zero by anything gives zero. So a machine built from only those six instructions is, quite literally, a machine that does nothing, no matter how sophisticated its circuitry looks.*

The seventh instruction, $\sigma^5 = Z$, is different: it adds one. *Addition moves the origin. That single instruction is what makes all* 81 *states reachable — and hence what makes the machine a computer instead of an ornament.*



Proved by exhaustive search, Pass 2774. Breadth-first search over all $3^4 = 81$ frames from $(0,0,0,0)$:

CX alone	1
all six symplectic opcodes	1
full ISA (adding $\sigma^5 = Z$)	81

Reproduce: `py -3 analysis/w33_pass2774_isa_frame_reachability.py`
Certificate: `data/PART_W33_PASS2774_ISA_FRAME_REACHABILITY.json`

5.4 How much computation the eight opcodes actually buy

In plain language. *"All 81 states are reachable" is weaker than it sounds — it says you can get anywhere, not that you can perform every operation. The stronger question is: what is the complete list of things this instruction set can do?*

Proved, Pass 2778. The six Clifford opcodes generate a group of order exactly 51,840 — that is, the whole of $\mathrm{Sp}(4,3)$, nothing missing. Adjoining the translation gives

$$\langle \mathcal{I}_{\mathrm{holo}} \rangle \;=\; \mathrm{ASp}(4,3) \;=\; \mathbb{F}_3^4 \rtimes \mathrm{Sp}(4,3), \qquad |\cdot| = 81 \times 51840 = 4{,}199{,}040.$$

One translation generator suffices. Enabling only Z on the past register gives the same group as enabling both, because $\mathrm{Sp}(4,3)$ is transitive on the 80 nonzero frame vectors: the orbit of a single translation already spans $\mathbb{F}_3^4$. Measured orbit size: 80. A second translation instruction adds nothing and can be removed from the ISA.

Eight opcodes. Three bits. 4,199,040 **distinct operations, exactly — and the eighth opcode is provably redundant.**

5.5 How small can the instruction set be?

In plain language. *If one opcode is redundant, how many are? This is not idle: every instruction removed is decoder logic removed, and if enough go, the opcode field itself gets shorter — which shortens every instruction word in the machine.*
The answer is that you can throw away half of them.

Exhaustive over all subsets, Pass 2789. Because $\mathbb{F}_3^4 \rtimes H = \mathrm{ASp}(4,3)$ iff $H = \mathrm{Sp}(4,3)$ and at least one translation is present, the search reduces to the six linear opcodes:

subsets of size 1 generating $\mathrm{Sp}(4,3)$:	0
subsets of size 2 :	0
subsets of size 3 :	**6**

The six minimal triples are $\{F_p, F_f, \mathrm{CX}_{p\to f}\}$, $\{F_p, F_f, \mathrm{CX}_{f\to p}\}$, $\{F_p, S_f, \mathrm{CX}_{p\to f}\}$, $\{F_p, \mathrm{CX}_{p\to f}, \mathrm{CX}_{f\to p}\}$, $\{F_f, S_p, \mathrm{CX}_{f\to p}\}$, $\{F_f, \mathrm{CX}_{p\to f}, \mathrm{CX}_{f\to p}\}$.

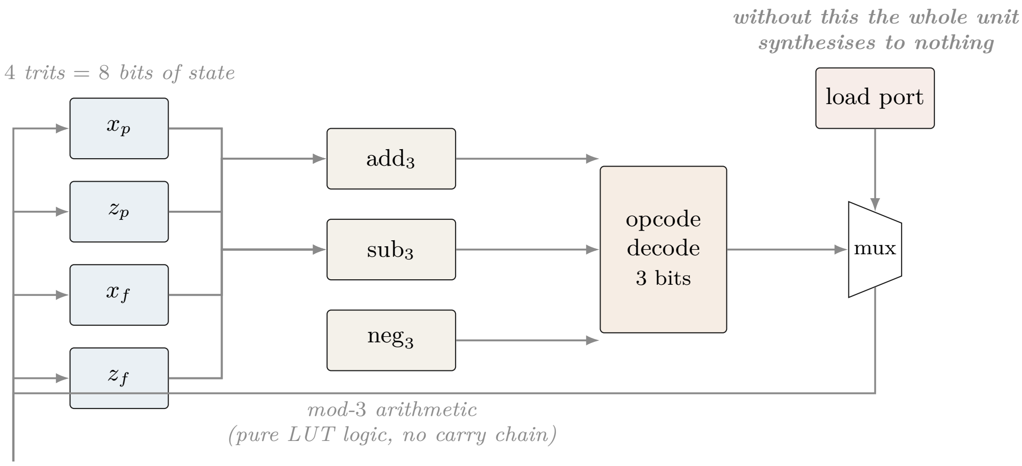
Three Clifford opcodes plus one translation generate everything. Four frame instructions — a two-bit opcode field, not three.

A trap avoided, and worth naming. A minimality *proof* says nothing about *area*. It is tempting to quote the 72-cell figure for the four-operation engine on the grounds that it does strictly less work — and that would be wrong twice over: the number was measured on a different design, and removing decode cases can *cost* cells when what was removed was being shared with what remains. The minimal engine was therefore synthesised and placed separately (§6.2), and only then quoted. It came out at 43 cells; had it come out at 80, that is the number this document would print.

In plain language. *Two details are worth noticing. First, there are* six *different minimal choices, so a chip designer has real freedom to pick whichever three are cheapest in silicon rather than being handed one answer.*
Second, every single one of the six contains at least one Fourier gate and *at least one controlled-add. No collection of phase gates and one Fourier gate will do. The entangler is not optional — which is the formal version of the intuition that a machine that never couples past to future is not really using the self-entanglement at all.*

6 The hardware

6.1 The frame unit datapath



`rtl/w33_pass2777_isa_frame_unit.sv`

6.2 The measured cell budget

In plain language. *Below is what the machine actually costs, measured by running the industry-standard open-source tools — Yosys to translate the design into logic gates, nextpnr to place those gates on a real chip — targeting a Lattice iCE40 UP5K, a chip you can buy on a breakout board for about \$25.*
The entire arithmetic core of this computer is seventy-two *logic cells, out of* 5,280 *available. It uses* 1.4% *of a \$25 chip.*

The three designs, on two parts, in one harness.

	HX8K CT256		UP5K SG48		
Design	**LC**	$F_{\max}$	**LC**	$F_{\max}$	**Opcode**
Minimal engine, 4 operations	**43**	**208.86**	43	72.40	2 bits
Public unit, 6 frame opcodes	72	175.72	72	60.80	3 bits
Engine + support readout	48	207.64	48	72.05	2 bits

frequencies in MHz

Same cells on both parts — synthesis does not know the package — but the HX8K is 2.9× faster. **Every UP5K figure elsewhere in this document therefore understates speed by about a factor of three**, and is kept because it is the part the $25 board carries.

Measured on UP5K SG48; the per-instruction breakdown.

Design	Cells	Pins	$F_{\max}$	Opcode
Minimal engine, 4 operations	**43**	22	**72.40** MHz	2 bits
Public unit, 6 frame opcodes	72	26	60.80 MHz	3 bits

40% **fewer cells and** 19% **faster.** The minimality theorem cashes out in silicon. Also verified on the minimal engine, because both checks are cheap and both have caught real defects here: the linear part is symplectic on frame *differences* (SAT, 4100 variables / 11278 clauses — a translation cancels in a difference, so one assertion covers all four opcodes at once), and no output folds to a constant.

Measured; iCE40 UP5K SG48, Yosys 0.67 + nextpnr.

Build	Logic cells	$F_{\max}$ (MHz)	Contains
F only	16	230.84	one Fourier gate
$F_p + F_f$	19	230.84	both Fourier gates
F and S (4 opcodes)	40	86.60	the single-qutrit Cliffords
CX only	33	72.40	the entangler, both directions
$\sigma^5{=}Z$ only	20	144.07	the translation
all Clifford, no Z	65	60.80	cannot leave the origin
full frame unit	**72**	**60.80**	the machine
For comparison, elsewhere in the design:			
CX, bare loadable	23	72.40	`w33_pass2752_cx_loadable_frame.sv`
36-lane mixer, serial	4048	19.65	`w33_pass2612_serial_mixer.sv`
36-lane mixer, parallel	13,965 LUT4 + 799 carry		does not fit any iCE40

In plain language. *The last two rows are worth a moment. The "mixer" is the part that combines* 36 *signals at once. Built the obvious way — all* 36 *lanes in parallel — it needs about* 14,000 *logic cells, nearly twice what the largest chip in this family has. Built serially, one lane at a time, it needs* 4,048*: about a third of the area, at* 1/36 *of the speed. That is the kind of trade a blueprint exists to make explicit.*

What we got wrong, and how we know.Every cell count in this project before Pass 2753 was wrong, and the reason is instructive. We reported 13 cells for F and S, 8 for CX, 42 for the ISA. Those modules had no load port. By the reachability theorem of §5, a register driven only by Clifford opcodes can never leave $(0,0,0,0)$ — so the synthesis tool *proved the state was constant and deleted six of the eight flip-flops.* We were measuring the area of nothing.

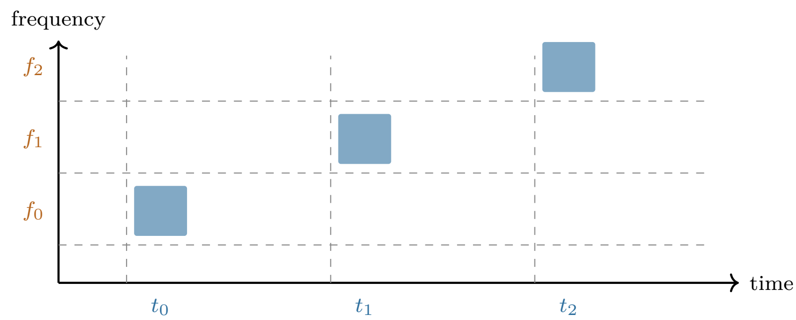
Two independent teams shipped this same defect within four hours of each other. Both had *exhaustive* testbenches — one covered all 81^2 pairs of frames — and both passed, because the testbenches drove the *combinational* logic and never instantiated the sequential wrapper.

Simulation cannot detect this class of defect at all. A module that folds to a constant still simulates correctly. It just does nothing. Only a synthesis frontend asks whether the state can move.

One figure survived unchanged: $\sigma^5 = Z$ was reported at 21 cells and re-measures at 20. It was always real — because it is the one instruction that is not linear, and so the one instruction that could always move the state. The theory predicted exactly which number would survive.

7 The physical layer

In plain language.*Everything above is logic — it would work on any hardware that can implement the eight operations. This section is about doing it with light, which is the intended implementation, and about what has already been demonstrated in a laboratory as opposed to designed on paper.*
A single photon carries several independent properties. Two of them are useful here: when *it arrives (its time bin) and* what colour *it is (its frequency bin). Chop time into three slots and colour into three bands, and one photon carries two qutrits — the "past" and "future" registers of §3.*



The sum gate
$|f,t\rangle \mapsto |f,\ t+f \bmod 3\rangle$
A chirped fibre Bragg grating delays each colour by a different amount, so the photon's *colour* controls how far its *arrival time* shifts. That is a controlled-add, implemented with one passive optical element.
Demonstrated: Imany *et al.*, *npj Quantum Information* **5**, 59 (2019). Three 3 ns time bins at 6 ns spacing, three frequency bins at 380 GHz, −2 ns/nm dispersion, 18 ns wraparound. Measured basis fidelity 0.92 ± 0.01; certified entanglement of formation $\geq 1.19 \pm 0.12$ ebits.

Evidence boundary — what is measured and what is not. Measured in a laboratory: the deterministic single-photon qutrit SUM gate, at fidelity 0.92 ± 0.01, with the entangled output certified.

Proved exactly: the group theory, the geometry, the class atlas, the ISA semantics, the reachability, the synthesis and placement figures.

Neither, and stated as such: source efficiency, insertion-loss engineering, fault-tolerance thresholds, and the magic-state injection route. The M_{36} resources are typed `M36_Q4_RAW` in the controller and injection is *refused* in RTL until a proved ququart protocol supplies a validity assertion. That refusal is a design feature: it makes the gap impossible to paper over.

8 Why the exponent nine

In plain language. *A small but pretty result that shows how tightly the layers fit. To tell which operation the machine just performed, you measure a quantity and look it up in a table. The obvious quantity — the "trace" — has a flaw: it changes if the whole state picks up an irrelevant overall phase, which physically means nothing. The fix used in this project is to raise the trace to the* ninth *power and divide by a determinant. The ninth power was found empirically. Where does nine come from?*

Derived, Pass 2779. For a d-dimensional representation, under $U \mapsto \lambda U$ we have $\mathrm{Tr}(U^k)^e \mapsto \lambda^{ke}\,\mathrm{Tr}(U^k)^e$ and $\det(U^k) \mapsto \lambda^{kd}\det(U^k)$, so

$$\Theta_k(g) = \frac{\mathrm{Tr}(U_g^k)^e}{\det(U_g^k)}$$

is phase-invariant for all λ *exactly* when $e = d$. Here $d = 9$, the dimension of the two-qutrit state space — which by §3 is $\dim \mathrm{End}(\mathbb{C}^3)$. Verified numerically: $e \in \{3, 8, 10\}$ fail, $e = 9$ holds.

If the phase ambiguity were a *finite* group μ_m one could use any $e \equiv 9 \pmod{m}$, possibly smaller. We computed the scalar subgroup of the generated matrix group by collision sampling and it is μ_{12}. Since $9 \not\equiv 0 \pmod{12}$, the determinant is genuinely required, and 9 is the *smallest* exponent that works: the sensor is minimal.

$\mu_{12} = \mu_4 \times \mu_3$. The μ_3 is ω, the qutrit alphabet. The μ_4 is the quadratic Gauss sum $\sum_j \omega^{j^2} = i\sqrt{3}$, which is *imaginary* precisely because $3 \equiv 3 \pmod 4$ — the same congruence that makes time reversal a physical rather than a gauge symmetry.

In plain language. *That last box is the kind of thing that makes this project worth doing. A practical question — "why is the exponent nine in the measurement circuit?" — turns out to have the same answer as an abstract one — "why does this geometry distinguish past from future?". Both are the statement that* -1 *is not a perfect square when you count modulo* 3.

8.1 And for wider registers, it is usually three

Pass 2791. The phase group is not an artefact of the two-qutrit case. At $n = 1$ the Clifford group is small enough to enumerate exactly rather than sample:

$$|\langle F, S, X\rangle| = 2592 = \underbrace{12}_{\mu_{12}} \times \underbrace{216}_{|\mathrm{ASp}(2,3)|} \qquad \text{— consistent, so } \mu_{12} \text{ again.}$$

For n qutrits $d = 3^n$, so the minimal exponent is the least $e \equiv 3^n \pmod{12}$, and $3^n \bmod 12$ cycles $3, 9, 3, 9, \ldots$:

n	1	2	3	4
$d = 3^n$	3	9	27	81
minimal e	**3**	9	**3**	9

Verified directly on 3×3 unitaries: $e = 3$ is phase-invariant, $e \in \{2, 4, 9\}$ is not.

Odd register widths need only a *cube* of the trace, not a ninth power — a strictly cheaper measurement, alternating with width.

9 The magic states, and the protocol that was found

In plain language. *Everything described so far is, in a precise technical sense,* not enough. *The eight opcodes generate a large and useful group, but a machine restricted to those operations can be simulated efficiently by an ordinary computer. To get a genuine quantum advantage you need one more ingredient, traditionally supplied as a special prepared state called a* magic state.

The substrate hands us 36 *candidates for free —* 36 *special directions in the four-dimensional space, called the Witting rays. The question is whether those* 36 *can be* purified*: whether noisy copies can be distilled into cleaner ones. Without that, the machine is a very elegant classical simulator.*

This section is the one that changed most between drafts, and the sequence is worth following, because it is a small case study in how a negative result should be read. First an exhaustive search said no. Then an argument from resource accounting said the "no" could not be fundamental. Then a wider search said yes.

But before any of that: the 36 *are not a lucky accident. Count them against the machine's own address space.*

9.1 Magic is everything except one line

In plain language. *The machine addresses* 40 *points. There are* 36 *magic rays. That leaves* 4 *— and a* line *of this geometry contains exactly* 4 *points. Either a coincidence, or the whole story.*

Add the four missing directions back — they turn out to be the four coordinate axes, which is to say the ordinary, unmagical computational basis — and ask what the 40 *look like together.*

Exact, Pass 2835. Adjoin $e_1, \dots, e_4$ to the 36 rays and compute the orthogonality graph on all 40:

$$\text{degrees } \{12{:}40\}, \qquad \lambda = 2, \qquad \mu = 4 \implies \mathrm{SRG}(40, 12, 2, 4),$$

which *is* the $W(3,3)$ collinearity graph of §4. The four axes are mutually orthogonal — a 4-clique, and in a generalised quadrangle of order $(3,3)$ the 4-cliques are exactly the lines. Every magic ray is orthogonal to exactly one axis ($\{1{:}36\}$), which is the defining GQ axiom, and the resulting nearest-point partition is $9+9+9+9$ — precisely the four families the ray construction produces.

M_{36} is the 40 points of $W(3,3)$ with *one line* deleted, and the deleted line is the computational basis.

In plain language. *This changes what the magic states* are. *They are not a resource bolted onto the substrate and supplied from outside. They are the generic condition of the machine's own address space: pick a point at random and it is magic with probability* 36/40. *The ordinary, classically-simulable states are the single distinguished line you have to delete to get rid of the magic.*

One more thing falls out, and it is a warning about reading structure off counts. The geometry partitions the 36 *into four families of nine. The Clifford group partitions them into* $4+8+12+12$. Neither refines the other. *The symmetry and the geometry are looking at different things.*

9.2 All thirty-six look identical from the inside

Exact integer arithmetic in $\mathbb{Z}[\omega]$, Pass 2790. Only two overlap values occur among the 36 rays:

$$|\langle i|j\rangle|^2 \in \{\,0,\ \tfrac{1}{3}\,\},$$

and every ray has the *same* profile: 11 orthogonal partners and 24 partners at overlap 1/3. There is exactly one Gram profile among all 36.

(The orthogonality graph is 11-regular on 36 vertices but *not* strongly regular: $\lambda \in \{1,2\}$, $\mu \in \{3,4\}$. Recorded because the near-miss invites a false claim.)

In plain language. *So no symmetry of the machine can tell one ray from another by looking at how they sit relative to each other. If the rays differ at all, they differ only in how they sit relative to the* ordinary *states — an external comparison, not an internal one.*

9.3 And from the outside, they fall into exactly three grades

60 **two-qubit stabilizer states,** $F_{\mathrm{stab}} = \max_s |\langle s|\psi\rangle|^2$**.**

F_{stab}	closed form	rays	grade
0.750000	$9/12$	**4**	shallow
0.705342	$(5+2\sqrt{3})/12$	**24**	mid
0.622008	$(4+2\sqrt{3})/12$	**8**	deep

Stabilizer fidelity is a Clifford invariant, so the $8+24+4$ grading is genuine and not an artefact of the chosen basis.

All three noise thresholds are one formula. For $\rho_p = (1-p)|m\rangle\langle m| + p\,I/4$ the target overlap is $1-3p/4$, so the witness certifies non-stabilizerness exactly while $1-3p/4 > F_{\mathrm{stab}}$:

$$\boxed{p \;<\; \tfrac{4}{3}\big(1-F_{\mathrm{stab}}\big)}$$

shallow	0.333333333333	=	$1/3$
mid	0.392877598318	=	$(7-2\sqrt{3})/9$
deep	0.503988709429	=	$(8-2\sqrt{3})/9$

Three separately-published thresholds, one formula, agreement to 10^{-12}.

9.4 Four, not three: fidelity is necessary but not sufficient

What we got wrong, and how we know. An earlier draft of this section said: *"the distillation search needs three representatives, not thirty-six."* That was one step too far. Equal stabilizer fidelity does not imply the two states are interchangeable — it is a single number, and a single number rarely separates everything.

Acting with the full two-qubit Clifford group (92,160 matrices) on the 36 rays and matching images by *fidelity* rather than by a rounded key:

$$\text{Clifford equivalence classes} = 4, \qquad \text{sizes } [4,\ 8,\ 12,\ 12].$$

Every class sits inside one grade — so the grading is real — but the 24-ray middle grade is **two** inequivalent families of twelve. The correct count is **four**.

Corroborated from the other direction: an independent exhaustive distillation search, which knows nothing about Clifford orbits, reports results for "*both* middle grades". Two unrelated computations, the same splitting.

And nothing an overlap can see tells the twins apart, Pass 2830. The *full* stabilizer overlap spectrum — the multiset of $|\langle s|\psi\rangle|^2$ over all 60 stabilizer states — is a Clifford invariant, and every stabilizer Rényi entropy and overlap moment is a function of it. Computed exactly in ninths, the two 12-ray classes have the **identical** spectrum.

Across all four classes there are only **three** distinct spectra. So the entire 60-value distribution has exactly the same resolving power as its single largest entry, F_{stab}.

Consequence. A protocol that reads its input only through overlap statistics *must* treat the two classes alike. The split can matter only to a protocol that uses the ray's Clifford orbit — one that fixes a frame and cares which representative it was handed.

9.5 Why the first "no" could not have been the last word

In plain language. *An exhaustive search over* 5,355 *small codes and* 21,420 *branches found no two-copy protocol that improves fidelity. That is a real result, and the tempting reading is that two noisy copies simply do not contain enough raw material to make one better copy.*
There is a way to check that reading directly, without searching any protocols at all. Physicists have quantities that measure *how much magic a state contains and which are guaranteed never to increase under any allowed operation — no measurement, no post-selection, no clever feedback can make one go up. If two copies scored lower than the target, distillation would be impossible for reasons no ingenuity could overcome. So: compute the score.*

$D_{\min} = -\log_2 F_{\text{stab}}$ **over all** 36,720 **four-qubit stabilizer states.**

grade	$F_{\text{stab}}(\psi)$	$F_{\text{stab}}(\psi^{\otimes 2})$	$D_{\min}(\psi)$	$D_{\min}(\psi^{\otimes 2})$
shallow	0.750000000	0.562500000	0.415037	0.830075
mid	0.705341801	0.497507057	0.503606	1.007211
deep	0.622008468	0.386894534	0.684994	1.369988

F_{stab} is *exactly* multiplicative here, so $D_{\min}(\psi^{\otimes 2}) = 2D_{\min}(\psi) > D_{\min}(\psi)$ for every grade. The budget is never the binding constraint.

The monotone never obstructs. Any two-copy no-go in this system is *structural*, not information-theoretic — a fact about the protocol family searched, not about the resource. The right response to it is to widen the family.

9.6 And widening the family worked

Deep-grade distillation, exact. Extending the search from one decoder to all 11,520 logical Clifford decoders gives 48 **improving branches** on the deep eight-ray grade, and zero on the shallow and both middle grades. One explicit branch — input ray 5, stabilizers $IYZY$ and $YZXY$, syndrome $(-1,+1)$, output ray 7 — has the exact curve

$$P_{\text{succ}} = \frac{p^2 - 2p + 2}{4}, \qquad F_{\text{out}} = \frac{5p^2 - 12p + 8}{4(p^2 - 2p + 2)},$$

$$F_{\text{out}} - F_{\text{in}} = \frac{p(p-1)(3p-2)}{4(p^2 - 2p + 2)} \; > \; 0 \quad \text{for } 0 < p < \tfrac{2}{3},$$

an interval that contains the whole deep-grade magic window $p < (8 - 2\sqrt{3})/9 \approx 0.504$.

In plain language.*So the resource is usable. Note the shape of the story: the first search was correct and its conclusion was correctly stated; the accounting argument said the obstruction could not be fundamental; the wider search then found the protocol. Nobody was wrong at any step — but a reader who had stopped at "exhaustive search found nothing" would have drawn exactly the wrong conclusion about the machine.*
And now the part that a reader should not *stop before.*

9.7 Usable is not the same as practical

The convergence rate, Pass 2831. Linearising the accepted-round map at the pure fixed point:

$$\left.\frac{dp'}{dp}\right|_{p=0} = \frac{2}{3}, \qquad P_{\text{succ}}\big|_{p=0} = \frac{1}{2}.$$

The noise is multiplied by two thirds each accepted round. It is **not** squared. Each round consumes two inputs and succeeds half the time, so the raw cost multiplies by $2/P_{\text{succ}} \approx 4$ per round while the infidelity only falls by a third.

start p	to 10^{-3}	to 10^{-6}	to 10^{-9}
0.10	2.3×10^{7}	4.0×10^{17}	6.9×10^{27}
0.20	5.6×10^{8}	9.7×10^{18}	1.7×10^{29}
0.30	1.5×10^{10}	2.6×10^{20}	4.5×10^{30}
0.50	1.1×10^{14}	1.9×10^{24}	3.2×10^{34}

raw noisy states consumed per distilled output

The map converges *linearly*. Standard distillation is quadratic or cubic — Reed–Muller 15-to-1 gives $p' \approx 35p^3$ — and that gap is the whole game.

In plain language. *So: the deep-grade branch is a genuine fidelity-improving map and a genuine refutation of the earlier no-go. It is* not*, by itself, a usable distillation routine. Ten to the twenty-seven raw states is not an engineering number.*

What it establishes is that the resource is not inert — which is the thing that was genuinely in doubt. Making it practical needs a branch with super-linear convergence, or a portfolio composing several of the 48*. This document reports the honest state of that: one improving map, quantified, and a clear statement of what would have to be found next.*

A near-miss in this very section. The first draft of the yield analysis described the cost as "doubly exponential in a good way — the infidelity squares each round." It does not; it falls by a factor of $2/3$. The tables above were already printed and would have looked like corroboration to anyone who did not check the ratio between successive rows. The linearisation is one derivative and it changes the conclusion from "deep targets are cheap" to "deep targets are impossible." Compute the rate; do not eyeball the trend.

Evidence boundary, stated as bluntly as possible. What the deep-grade result *is*: an exact state-fidelity distillation theorem for an explicitly enumerated class — all 5,355 binary $[[4, 2]]$ stabilizer projectors, all four syndromes, all 11,520 logical Clifford decoders.

What it is **not**: a fault-tolerant injection threshold, an asymptotic-yield theorem, or a laboratory demonstration. The controller still types the resource `M36_Q4_RAW` and refuses injection in RTL until a proved *fault-tolerant* protocol asserts validity, because a fidelity-improving map is not the same object as an injection threshold. **Universality is now a matter of closing a stated gap rather than of finding an unknown protocol** — which is a different and much better position than the previous draft of this document described.

10 Support for readout, phase for execution

In plain language. *Here is a design temptation, and the theorem that kills it.*

The machine's frame is four trits — 81 *states. But a great deal of what you want to* observe *about the frame is coarser than that: often you only care* which *of the four registers are non-zero, not what they hold. That is a four-bit "support mask",* 16 *possibilities instead of* 81*, and it has beautiful structure — it forms an exact finite geometry with the capacity profile*

$$(4, 6, 4, 1) \odot (1, 2, 4, 8) = (4, 12, 16, 8).$$

A five-fold compression with geometry attached is exactly the kind of thing an engineer wants to build the machine out of. Can the machine simply run *on support masks and skip the phases?*

No. And the counterexample is two lines long.

The obstruction. The two frames

$$(0,1,0,0) \qquad \text{and} \qquad (0,2,0,0)$$

have the same support mask `0100`. Apply Z_p — the translation, $z_p \mapsto z_p + 1$ — and their masks become `0100` and `0000`. Two states that the support layer cannot tell apart are sent to two states it *can.*

So support is not a *congruence*: it is not preserved by the dynamics, and a machine that stored only support would be unable to predict its own next state. Refining the partition until it is stable gives

$$\boxed{16 \longrightarrow 40 \longrightarrow 78 \longrightarrow 81}$$

with class-size histograms $\{1{:}1, 2{:}4, 4{:}6, 8{:}4, 16{:}1\} \to \{1{:}7, 2{:}29, 4{:}4\} \to \{1{:}75, 2{:}3\} \to \{1{:}81\}$. The terminal partition is *discrete*: full ternary phase is forced.

Support for readout, phase for execution. The compression is real and exact — but it is a lens, not a substrate.

10.1 What a readout costs, in joules

In plain language. *A support readout throws information away — that is what "lossy" means. Landauer's principle says throwing information away has an unavoidable energy cost, and here the cost is exactly computable, because the fibres are exactly known: a mask of weight k has exactly 2^k preimages, and there are $\binom{4}{k}$ such masks.*

Exact, Pass 2836.

$$H(\text{state} \mid \text{support}) = \frac{1}{81}\sum_{k=0}^{4}\binom{4}{k}2^k\,k = \frac{216}{81} = \boxed{\tfrac{8}{3}\ \text{bits}}$$

$$H(\text{state}) = \log_2 81 = 6.339850, \qquad H(\text{support}) = 3.673183, \qquad \frac{3.673183}{4} = 91.83\%\ \text{efficient.}$$

Over $\mathbb{F}_q$ the closed form is $4(q-1)\log_2(q-1)/q$, verified exactly at $q = 2, 3, 4, 5, 7, 8, 9, 11$.

In plain language. *At $q = 2$ that formula gives* zero. *Over a two-element field the support* is *the state, so a support readout destroys nothing at all. Every one of the 8/3 bits is the price of the third field element.*

That is the thermodynamic version of the architectural law: ***the phase is precisely the part you pay to look at.***

Landauer, at room temperature.

$$E \;\geq\; \tfrac{8}{3}\,k_B T \ln 2 \;=\; 7.656 \times 10^{-21}\,\text{J} \;=\; 47.78\ \text{meV} \quad \text{at } 300\,\text{K},$$

or 7.66 pW of unavoidable dissipation at a 1 GHz readout rate. This is a floor set by information theory, independent of the circuit: no implementation of the support readout can beat it.

And the counting floor lands exactly where the search does. $\lceil \log_2 81 \rceil = 7$ **bits are needed to label** 81 **states; an exhaustive search shows no** 7 **support taps suffice and exactly** 48 **eight-tap sets do. The observer is one bit above the floor, and that bit is provably necessary.**

10.2 Enforcing the law in the netlist

In plain language. *The law is a theorem about mathematics. It constrains no circuit. Nothing stops an engineer from wiring the cheap* 4*-bit mask back into the execution path — and the resulting machine would pass simulation almost always, because support* is *preserved by most operations. It would drift the first time a translation fired on a register holding a* 2*.*

That is the same failure shape as the folded registers of §6.2: correct in simulation, wrong in silicon, invisible to the testbench. So the law is enforced structurally instead of documented.

Gate-level information flow, Pass 2834. `scripts/check_information_flow.py` flattens the design to gates, builds the driver graph and computes reachability in both directions:

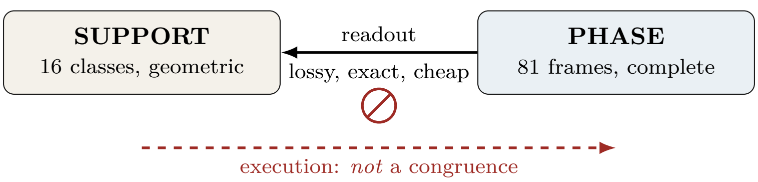
```
top w33_support_readout_diode: 59 cells after flatten+opt
reverse: support_mask, tamper_mask reach none of xp_o, zp_o, xf_o, zf_o
forward: every frame register reaches the mask
PASS: the readout is a diode.
```

Self-tested against a negative control — rewiring a tamper input into the load enable makes it report VIOLATION on all four registers — because a guard that can only pass is not a guard. The non-congruence witness is asserted in the netlist too and SAT-proved (228 variables, 600 clauses).

This is the one check in this project that **fails** rather than warns. Unlike a rediscovery collision, a backward edge has no benign reading.

In plain language. *It is also nearly free. Adding the readout to the minimal engine costs* ***five logic cells*** *and* 0.6% *of the clock:* $43 \to 48$ *cells,* $208.86 \to 207.64$ *MHz. The cheap layer really is cheap — and now it is cheap* and *provably one-way.*

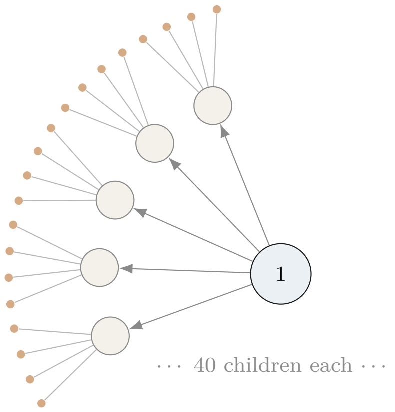
In plain language. *This is a useful kind of negative result, because it tells you exactly where to put the cheap representation. Readout, routing, telemetry and visualisation can all live in the* 16*-class support world and enjoy its geometry. The execution core cannot. A machine built the other way round would appear to work in simulation — support is preserved by* most *operations — and would drift silently the first time a translation fired on a register holding a* 2*.*



11 The network

11.1 Fractal scaling

In plain language. *The machine is designed to be copied. A node contains 40 sub-nodes; each sub-node contains 40 of its own; and so on. The network of machines has the same shape as the inside of one machine — which means one routing scheme, one addressing scheme, and one error model serve every scale at once.*



$N_n = 40^n$ **leaves** at depth n

$I_n = \dfrac{40^n - 1}{39}$ **instances** in total

routing depth $= 8n = 8\log_{40} N$

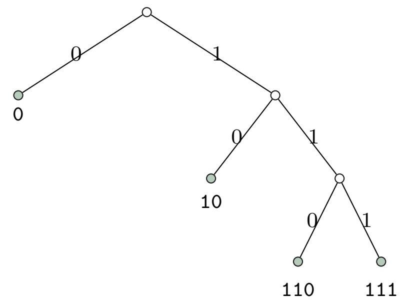
where $8 = 3 + 5$: three trits of local address plus five of the spread index. Address length is logarithmic in the size of the entire network, at every scale, by construction.

11.2 Every bit string is a legal address

Exactly. The routing code has orbit sizes $162 + 162 + 324 + 648 = 1296$ on compass pairs, giving codeword lengths $3, 3, 2, 1$ and

$$2^{-3} + 2^{-3} + 2^{-2} + 2^{-1} \;=\; 1 \quad \text{exactly (Kraft equality).}$$

The prefix code is $\{110, 111, 10, 0\}$ with expected length $7/4$. Because Kraft holds with *equality*, the decision tree has no unused branch: 20,000 random bit strings were parsed with 0 failures.



In plain language. *Look at the tree: every branch ends in a valid address. Nothing dangles. The engineering consequence is unusual and worth stating plainly. In a normal computer, a corrupted instruction stream can produce an* illegal instruction *— a bit pattern that means nothing, which the processor must detect and trap. Here that cannot happen. A corrupted packet is always a legal packet addressed somewhere else. The fault model therefore has no "decode error" term at all; it has only a "wrong result" term. That simplifies the error correction enormously, and it is forced by the geometry rather than designed in.*

What we got wrong, and how we know. We wrote that "the consequence nobody has stated" was that every bit string is a valid instruction stream. The project's own holonet manuscript says it at line 382 — "the routing words fill a binary decision tree with *no unused branch*" — with a citation to McMillan. This was the first error in this project caught by an automated guard rather than by a human noticing. The engineering consequence drawn on top (no decode-error term in the fault model) is a step past the coding fact and appears to be new; the completeness property itself is the manuscript's.

11.3 Remote operations

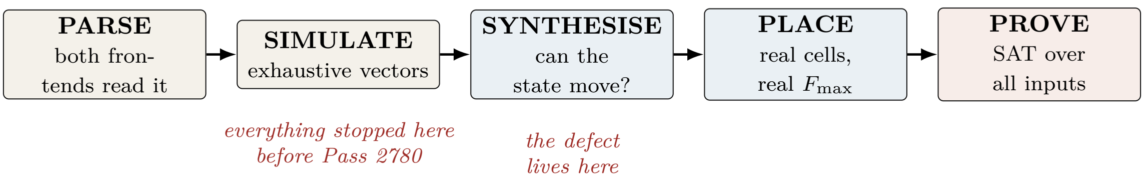
Exactly. A SUM gate between two *separate* machines costs exactly:

$$\boxed{\text{one shared qutrit Bell pair} \;+\; \text{two classical trits}}$$

Protocol: Alice applies reverse local SUM into her half of the pair, measures, sends trit m; Bob applies $X^{-m}F^2$, applies direct SUM to his data, measures in the Fourier basis, sends trit n; Alice records Z^n in her Pauli frame. All nine branches implement $|x, y\rangle \mapsto |x, y + x\rangle$; verified on all nine basis inputs, all nine branches, and 32 random superpositions. Total conditional success probability 1.

12 Verification: how we know

In plain language. *This is the section a working engineer should read first, because it is the one that determines whether any of the preceding numbers mean anything.*
The short version: this project has repeatedly shipped results that were verified in the direction that would confirm them and never in the direction that would break them. Every tool below exists because that happened at least three times.



The five automated guards, each built from a measured failure.

Guard	Catches	Built because
`check_rediscovery.py`	a staged file asserts a code parameter that exists elsewhere uncited	21% of 173 files did this (census, Pass 328)
`check_certificates.py`	a frozen certificate that cannot reproduce its own digest	one was unverifiable *from birth* and passed every check while being so
`check_novelty_claims.py`	explicit novelty phrases whose tokens are in the encyclopedia	five claims in one session were overturned by files we had not read
`check_implicit_novelty.py`	*emphasised* results carried by a source the file never cites	four of six failures never used a novelty phrase at all
`check_rtl_folds.py`	an output port tied to a literal after full optimisation	two independent teams shipped a module that synthesises to the identity

All five **warn and never block**. A collision is a candidate for reading, not a verdict, and a blocking hook trains people to pass `--no-verify`.

What we got wrong, and how we know.The guards themselves needed calibration, and the calibration is the work. `check_implicit_novelty.py` went through four measured versions:

Flagged	Version	What the noise turned out to be
9/122	presence of any token	all nine were *pass numbers in titles*
0/122	+ skip files citing “Prior art”	vacuous — every file has that section
51/122	+ cite the *right* source per token	`W(3,3)` 21×: the repo’s own subject
27/122	+ rarity and non-round integers	what ships

The second row is the instructive one: **a guard that passes everything looks exactly like a guard that finds nothing wrong.** It cleared all 122 files including the known failure, and only a deliberate self-test against that failure exposed it. Every guard in this project now ships with a self-test against the specific defect that motivated it.

`check_rtl_folds.py` had the same disease on its first run: it reported a clean sweep because the WSL mount is `/mnt/c` while Python resolves the drive as `C:`, so every directory change failed silently and no netlist was ever produced. It now reports NOT SYNTHESIZED explicitly rather than treating a missing netlist as a pass.

13 The complete ledger

In plain language.*Everything this blueprint asserts, with how it is known. If a row says “proved” there is a machine-checkable artifact; if it says “measured” there is a tool output; if it says “published” someone else did it and we cite them.*

Claim	**Status**	**Artifact**
$W(3,3)$ has $\mathrm{SRG}(40,12,2,4)$ collinearity graph	classical	literature
$40 = (3^4 - 1)/2$ from one self-entangled qutrit	proved	Pass 2724
The one-qutrit thesis itself	**prior art**	`index.html` MCLXVII
Outer involution = transpose = time reversal	proved at q=3	Pass 2732
Time reversal outer $\iff$ $q \equiv 3 \pmod 4$	proved	Pass 2750
Transpose *constructed* and checked at $q \leq 23$	proved	Pass 2792, 8 primes
Anti-symplectic T, $T^{\top} J T = -J$; class action 10 pairs + 14 fixed	proved	parallel track, Pass 2762
Six Clifford opcodes reach 1 frame of 81	proved	Pass 2774 (BFS)
Full ISA reaches all 81	proved	Pass 2774
Clifford opcodes generate $\mathrm{Sp}(4,3)$, order 51,840	proved	Pass 2778
Full ISA generates $\mathrm{ASp}(4,3)$, order 4,199,040	proved	Pass 2778
One translation generator suffices (orbit = 80)	proved	Pass 2778
Minimal Clifford subset has size 3; sizes 1, 2 give none	proved	Pass 2789, 6 triples
Frame ISA fits a **two-bit** opcode field	follows	Pass 2789
Sensor exponent 9 = dim; phase group μ_{12}; 9 minimal	proved	Pass 2779
μ_4 factor = imaginary Gauss sum	proved	Pass 2779
μ_{12} at n=1 by exact enumeration ($2592 = 12 \cdot 216$)	proved	Pass 2791
Minimal exponent = 3^n mod 12: 3 odd, 9 even	derived	Pass 2791
36 rays share one Gram profile (11 orth., 24 at 1/3)	proved	Pass 2790, $\mathbb{Z}[\omega]$
8+24+4 is the stabilizer-fidelity spectrum	proved	Pass 2790, 60 stab. states
All three witness thresholds $= \frac{4}{3}(1 - F_{\mathrm{stab}})$	derived	Pass 2790, 10^{-12}
Clifford classes on the 36 rays: $[4,8,12,12]$	proved	Pass 2797
the 24-ray grade splits; "three" was wrong	**corrected**	Pass 2797
F_{stab} exactly multiplicative on two copies	measured	Pass 2798, 36,720 states
The monotone never forbids two-copy distillation	proved	Pass 2798
Deep grade: 48 improving branches, $0 < p < 2/3$	**prior art**	parallel track, Pass 2821
Support is not an execution congruence: 16→40→78→81	**prior art**	parallel track, Pass 2822
Phase group = μ_{12} for *every* n	proved	Pass 2799
Minimal engine: **43** LC, 72.40 MHz (UP5K)	**measured**	Pass 2796
same engine on HX8K: **208.86** MHz	**measured**	Pass 2833
M_{36} is $W(3,3)$ minus one line	**proved**	Pass 2835, $\mathrm{SRG}(40,12,2,4)$
the deleted line is the computational basis	proved	Pass 2835
Support readout erases exactly 8/3 bits	**proved**	Pass 2836
$4(q-1)\log_2(q-1)/q$; zero at q=2	proved	8 field sizes
Landauer 47.78 meV at 300 K	derived	Pass 2836
The two 12-ray classes share the full spectrum	**proved**	Pass 2830
Deep-grade map is *linearly* convergent, rate 2/3	**proved**	Pass 2831
$\sim 10^{27}$ raw states for 10^{-9} infidelity	computed	Pass 2831
Three-copy super-multiplicativity	no witness	not a proof either way
Readout diode: mask reaches no frame register	**proved**	gate level, Pass 2834
symplectic on differences, all four opcodes	SAT	4100 vars / 11278 clauses
Public frame unit: 72 LC, 60.80 MHz	**measured**	§6.2
Loadable CX: 23 LC, 72.40 MHz, CX^3=I	measured + proved	5265 vars / 14005 clauses
Parallel 36-mixer: 13,965 LUT4, does not fit	measured	Pass 2773
Signed mixer correct on impulse class	proved	1,097,152 vars
Kraft equality; 0 parse failures in 20,000	proved + measured	Pass 2682
Photonic qutrit SUM, fidelity 0.92 ± 0.01	**published**	Imany *et al.* 2019
Remote SUM: 1 Bell pair + 2 trits	proved	parallel track, Pass 2771
M_{36} fault-tolerant *injection* threshold	OPEN	refused in RTL
Asymptotic distillation yield	OPEN	fidelity theorem only
Do the two 12-ray classes differ operationally?	OPEN	Clifford class only
Fault-tolerance threshold end to end	OPEN	—
Physical power in watts	OPEN	activity proxy only

14 Reproducing everything in this document

Commands, in order.

```
# reachability, the group, and the minimal generating sets
py -3 analysis/w33_pass2774_isa_frame_reachability.py
py -3 analysis/w33_pass2778_2779_affine_group_and_sensor_exponent.py
py -3 analysis/w33_pass2789_2792_minimal_isa_sensor_and_transpose.py

# the magic states: geometry, Clifford classes, and the monotone
py -3 analysis/w33_pass2790_witting_ray_geometry.py
py -3 analysis/w33_pass2797_2799_magic_orbits_and_monotone.py

# the minimal engine (43 LC) against the public unit (72 LC)
yosys -p "read_verilog -sv rtl/w33_pass2796_minimal_frame_engine.sv; \
  synth_ice40 -top w33_minimal_frame_engine -json m.json"
nextpnr-ice40 --up5k --package sg48 --json m.json --asc m.asc --freq 40

# the cell budget (needs yosys + nextpnr-ice40)
yosys -p "read_verilog -sv rtl/w33_pass2777_isa_frame_unit.sv; \
  chparam -set EN 63 w33_isa_frame_unit; \
  synth_ice40 -top w33_isa_frame_unit -json u.json"
nextpnr-ice40 --up5k --package sg48 --json u.json --asc u.asc --freq 50

# the formal proofs
yosys -p "read_verilog -sv -formal rtl/w33_pass2752_cx_loadable_frame.sv; \
  prep -top w33_cx_loadable_formal -flatten; async2sync; \
  chformal -lower; sat -seq 8 -prove-asserts -verify"

# the guards
py -3 scripts/check_rtl_folds.py
py -3 scripts/check_implicit_novelty.py
py -3 scripts/check_novelty_claims.py

# this document
tectonic holonet_machine_blueprint.tex
```

15 What is not built

In plain language. *A blueprint that lists only what works is advertising. Here is the honest remainder.*

1. **Two of the eight opcodes are not frame operations and are not built as such.** D_{12}-mirror is a transport operation in a dihedral group; M_{36}-magic is deliberately a *handshake* to an external state-preparation block, and the controller *refuses* injection until a proved protocol asserts validity. No non-Clifford unitary is invented in RTL, because inventing one would conceal the real gap.

2. **The universality gap moved but did not close.** A previous draft said no distillation protocol was known; §9 now gives an exact fidelity-improving one on the deep grade, valid for $0 < p < 2/3$. What remains open is narrower and better-defined: a fault-tolerant *injection* threshold and an asymptotic *yield*. A map that improves fidelity is not either of those, and the controller still refuses injection in RTL until one exists.

3. **No physical power figure exists.** The switching census gives about 1.622 output transitions per operation. That is a technology-independent activity proxy. Watts require the placed netlist, a capacitance model, voltage, clock and a representative workload.

4. **(Closed at Pass 2792.)** The transpose construction was previously verified at $q = 3$ only. It is now built and checked over $\mathbb{F}_q$ at $q = 3, 5, 7, 11, 13, 17, 19, 23$: $T^2 = I$ and $T^\top JT = -J$ hold at every one, and the congruence law holds at every one. Left in this list, struck through, because a blueprint should show which gaps closed and when.

5. **One original module is unparseable** — `rtl/w33_spread_mixer36.sv`, rejected by both frontends. It now carries a deprecation banner naming its replacement. It has since been removed from the tree entirely by the other track, which is the right call: a file no frontend can read is not a historical record, it is a trap for the next reader.

> *“Before recording a result, ask what single computation would falsify it, and run* that *one.* **maximal***,* **unique***,* **only***,* **exactly***,* **is** *— each of those words has a cheap negation.”*
>
> — the operating rule of this project, learned expensively